

# THE STRONGLY SUMMING DESIRED FAMILY IS NOT MINIMAL

FA\_BANACH.001, AGENT\_LANE.07

## STATUS AND SCOPE

Pellegrino and Santos ask in Problem 7.2 of [1] whether the family  $(\mathcal{L}_{ss,p})_{p \geq 1}$  of strongly  $p$ -summing multilinear operators is a maximal or minimal desired family. This packet gives a complete negative answer to the *minimality* half. The maximality half is not addressed, so the packet is classified as a partial solution to the source problem.

## SOURCE PASSAGE

Problem 7.2 occurs on PDF page 25 of [1]. The image below is a crop of the original arXiv PDF and includes the neighboring open problems.

The other properties are easily verified and hence  $(\mathcal{I}_p(E_1, \dots, E_n; F), \|\cdot\|_{\mathcal{I}_p})$  is a normed space for all  $E_1, \dots, E_n, F$ . The rest of the proof follows the lines of the previous proof.  $\square$

## 7. OPEN PROBLEMS

From the previous section two open problems arise:

**Problem 7.1.** *Is  $(\mathcal{L}_{sm,p})_{p \geq 1}$  a desired ideal? (we conjecture that it is not) If the answer is positive, it will be maximal.*

**Problem 7.2.** *Is  $(\mathcal{L}_{ss,p})_{p \geq 1}$  a maximal or minimal desired ideal?*

The answer to the next problem seems to be “NO”, but to the best of our knowledge, it is an open problem:

**Problem 7.3.** *Is  $(\mathcal{L}_{ss,p})_{p \geq 1} = (\mathcal{L}_{sm,p})_{p \geq 1}$ ?*

The exact question is:

**Problem 7.2.** Is  $(\mathcal{L}_{ss,p})_{p \geq 1}$  a maximal or minimal desired ideal?

## DEFINITIONS FROM THE SOURCE

For Banach spaces  $E_1, \dots, E_n, F$  and  $p \geq 1$ , an  $n$ -linear map  $T : E_1 \times \dots \times E_n \rightarrow F$  is strongly  $p$ -summing if

$$(1) \quad \left( \sum_{i=1}^m \|T(x_i^{(1)}, \dots, x_i^{(n)})\|^p \right)^{1/p} \leq C \sup_{\varphi \in B_{\mathcal{L}(E_1, \dots, E_n; \mathbb{K})}} \left( \sum_{i=1}^m |\varphi(x_i^{(1)}, \dots, x_i^{(n)})|^p \right)^{1/p}.$$

The least  $C$  is denoted by  $\|T\|_{ss,p}$ . Strongly multiple  $p$ -summability is the corresponding inequality on full Cartesian arrays; the source proves the norm-one inclusion  $\mathcal{L}_{ss,p} \subseteq \mathcal{L}_{sm,p}$ .

Write  $\mathcal{L}_{d,p}$  for the Banach multi-ideal of  $p$ -dominated maps. In the notation of [1], a family  $(\mathcal{I}_p)_{p \geq 1}$  is *desired* when:

- (1)  $\mathcal{L}_{d,p} \subseteq \mathcal{I}_p \subseteq \mathcal{L}_{sm,p}$  with both inclusions of norm at most one;
- (2) every  $\mathcal{I}_p$  is closed for scalar multiplication (csm) and under differentiation (cud);
- (3)  $\mathcal{I}_p \subseteq \mathcal{I}_q$  for  $p \leq q$ ; and
- (4) the multilinear Grothendieck and Dvoretzky–Rogers theorems hold.

A desired family is minimal if every desired family contractively contained in it is equal to it at every  $p$ .

## PROOF INTUITION

The Grothendieck requirement only forces a desired family to contain all maps  $(\ell_1)^n \rightarrow \ell_2$  at  $p = 1$ . We put in exactly the Banach multi-ideal generated by these maps, add the mandatory dominated ideal, and no more. The generated part is given an atomic norm whose central cost is the  $p$ -summing norm of the linearization. This makes completeness and all norm-one inclusions transparent.

To see that the resulting family is smaller than the strongly summing family, test trilinear forms on  $(e_i, e_i, e_i)$  in  $(\ell_2)^3$ . Every factorization through  $\ell_1$  is diagonal-null by Pitt’s compactness theorem, and the same is true of every dominated trilinear form by an exact finite-dimensional weak- $p$  calculation. The diagonal form  $\Delta(x, y, z) = \sum_i x_i y_i z_i$  is not diagonal-null, while scalar-valued forms are automatically strongly  $p$ -summing.

## THE ATOMIC GROTHENDIECK HULL

We use  $\pi_p$  for the absolutely  $p$ -summing norm of a linear operator. For  $A \in \mathcal{L}({}^n \ell_1; \ell_2)$ , let

$$\tilde{A} : \underbrace{\ell_1 \hat{\otimes}_\pi \dots \hat{\otimes}_\pi \ell_1}_{n \text{ factors}} \longrightarrow \ell_2$$

be its linearization. The domain is canonically isometric to  $\ell_1(\mathbb{N}^n)$ . Grothendieck’s  $\ell_1$ – $\ell_2$  theorem therefore gives

$$(2) \quad \pi_1(\tilde{A}) < \infty, \quad \pi_q(\tilde{A}) \leq \pi_p(\tilde{A}) \quad (1 \leq p \leq q < \infty).$$

Fix  $p, n, E_1, \dots, E_n, F$ . A  $p$ -atom is a map

$$(3) \quad S = u \circ A \circ (B_1, \dots, B_n),$$

where  $B_j : E_j \rightarrow \ell_1$ ,  $A : {}^n\ell_1 \rightarrow \ell_2$ , and  $u : \ell_2 \rightarrow F$  are bounded, and

$$(4) \quad \|u\| \pi_p(\tilde{A}) \prod_{j=1}^n \|B_j\| \leq 1.$$

Let  $\mathcal{A}_p(E_1, \dots, E_n; F)$  be the set of all such atoms. Define the synthesis operator

$$Q_p : \ell_1(\mathcal{A}_p(E_1, \dots, E_n; F)) \longrightarrow \mathcal{L}_{ss,p}(E_1, \dots, E_n; F), \quad Q_p(c) = \sum_S c_S S.$$

**Lemma 1.** *The synthesis map is well-defined and contractive. Its range*

$$\mathcal{G}_p(E_1, \dots, E_n; F) := Q_p(\ell_1(\mathcal{A}_p))$$

*is a Banach space for the quotient norm*

$$\gamma_p(T) = \inf\{\|c\|_1 : Q_p(c) = T\},$$

*and  $\|T\|_{ss,p} \leq \gamma_p(T)$ . As the spaces vary,  $(\mathcal{G}_p, \gamma_p)$  is a Banach multi-ideal. Moreover,  $\mathcal{G}_p$  is csm and cud, and  $\mathcal{G}_p \subseteq \mathcal{G}_q$  contractively whenever  $p \leq q$ .*

*Proof.* For one factorization (3), put  $y_i = B_1 x_i^{(1)} \otimes \dots \otimes B_n x_i^{(n)}$ . Absolute  $p$ -summability of  $\tilde{A}$  gives

$$\begin{aligned} & \left( \sum_i \|uA(B_1 x_i^{(1)}, \dots, B_n x_i^{(n)})\|^p \right)^{1/p} \\ & \leq \|u\| \pi_p(\tilde{A}) \sup_{\psi \in B_{(\ell_1 \hat{\otimes} \pi \dots \hat{\otimes} \pi \ell_1)^*}} \left( \sum_i |\psi(y_i)|^p \right)^{1/p} \\ & \leq \|u\| \pi_p(\tilde{A}) \prod_j \|B_j\| \sup_{\varphi \in B_{\mathcal{L}(E_1, \dots, E_n; \mathbb{K})}} \left( \sum_i |\varphi(x_i^{(1)}, \dots, x_i^{(n)})|^p \right)^{1/p}. \end{aligned}$$

Thus every  $p$ -atom has strongly summing norm at most one. Since  $\mathcal{L}_{ss,p}$  is Banach,  $Q_p$  is well-defined and contractive. Its kernel is closed, so  $\ell_1(\mathcal{A}_p)/\ker Q_p$ , identified with the range and equipped with the quotient norm, is Banach. This also proves  $\|T\|_{ss,p} \leq \gamma_p(T)$ .

Pre- and postcomposition in (3) gives the multi-ideal property and its norm inequality. Finite-type maps belong to the range: for a scalar monomial use  $B_j x = \varphi_j(x) e_1$ , the central coordinate-product map  $(z_1, \dots, z_n) \mapsto z_1(1) \dots z_n(1) e_1$ , and a rank-one postcomposition. For the canonical monomial on  $\mathbb{K}^n$  this representation has cost one, while its strongly summing norm is one; hence its  $\gamma_p$  norm is exactly one.

For csm, given  $\phi \in E_{n+1}^*$ , append  $B_{n+1} x = \phi(x) e_1$  and replace  $A$  by

$$A'(z_1, \dots, z_n, w) = e_1^*(w) A(z_1, \dots, z_n).$$

The linearization of  $A'$  is  $\tilde{A}$  after a norm-one tensor contraction, so  $\pi_p(\tilde{A}') \leq \pi_p(\tilde{A})$ . Applying this termwise to an atomic  $\ell_1$  expansion proves csm.

For cud, fix all variables except the  $j$ th. In each atom, partially evaluate  $A$  at  $B_k a_k$ ,  $k \neq j$ . The resulting central linear map is  $\tilde{A}$  after precomposition by

$$z \mapsto B_1 a_1 \otimes \cdots \otimes z \otimes \cdots \otimes B_n a_n,$$

whose norm is at most  $\prod_{k \neq j} \|B_k a_k\|$ . The ideal property of  $\pi_p$  again preserves summability of the atomic expansion.

Finally, (2) says that every  $p$ -atom is a  $q$ -atom when  $p \leq q$ . Hence  $\gamma_q(T) \leq \gamma_p(T)$  and the inclusion is contractive.  $\square$

#### A PROPER DESIRED SUBFAMILY

For each  $p \geq 1$ , define the componentwise sum

$$(5) \quad \mathcal{M}_p = \mathcal{L}_{d,p} + \mathcal{G}_p, \quad \|T\|_{\mathcal{M}_p} = \inf_{T=D+G} (\|D\|_{d,p} + \gamma_p(G)).$$

**Proposition 2.** *The family  $(\mathcal{M}_p)_{p \geq 1}$  is desired. For every  $p$ ,*

$$\mathcal{L}_{d,p} \xrightarrow{\|\cdot\| \leq 1} \mathcal{M}_p \xrightarrow{\|\cdot\| \leq 1} \mathcal{L}_{ss,p} \xrightarrow{\|\cdot\| \leq 1} \mathcal{L}_{sm,p}.$$

*Proof.* The source records the norm-one inclusion  $\mathcal{L}_{d,p} \subseteq \mathcal{L}_{ss,p}$ . Lemma 1 gives the same estimate for  $\mathcal{G}_p$ . Therefore the addition map

$$\mathcal{L}_{d,p}(E_1, \dots, E_n; F) \oplus_1 \mathcal{G}_p(E_1, \dots, E_n; F) \longrightarrow \mathcal{L}_{ss,p}(E_1, \dots, E_n; F)$$

is contractive. Its kernel is closed; consequently its range, equipped with the quotient norm (5), is Banach. It is a Banach multi-ideal, and the first two displayed inclusions are contractive. The last is the standard norm-one inclusion from [1]. The normalization of the multi-ideal norm follows from the cost-one representation of the canonical scalar monomial and the lower bound supplied by its strongly summing norm.

The  $p$ -dominated family is csm and cud and satisfies the inclusion theorem, as recalled in [1]; Lemma 1 gives the same properties for  $\mathcal{G}_p$ . They pass to the quotient sum, and in fact  $\|T\|_{\mathcal{M}_q} \leq \|T\|_{\mathcal{M}_p}$  for  $p \leq q$ .

It remains to check the coincidence axiom. If  $A : {}^n \ell_1 \rightarrow \ell_2$  is arbitrary, then (2) makes  $A$  one of the admissible central generators of  $\mathcal{G}_1$  (after a scalar normalization). Hence every such  $A$  belongs to  $\mathcal{M}_1$ : the required multilinear Grothendieck theorem holds. The source observes that any family sandwiched between  $\mathcal{L}_{d,p}$  and  $\mathcal{L}_{sm,p}$  has the required Dvoretzky–Rogers theorem. Thus all desired-family axioms hold.  $\square$

**Lemma 3.** *For every  $p \geq 1$  and every  $T \in \mathcal{M}_p(\ell_2, \ell_2, \ell_2; \mathbb{K})$ ,*

$$T(e_i, e_i, e_i) \longrightarrow 0.$$

*Proof.* First let  $G \in \mathcal{G}_p(\ell_2, \ell_2, \ell_2; \mathbb{K})$ . In every atom, each  $B_j : \ell_2 \rightarrow \ell_1$  is compact by Pitt's theorem [2]. Since  $e_i \rightarrow 0$  weakly in  $\ell_2$ , one has  $\|B_j e_i\|_1 \rightarrow 0$ . Bounded trilinearity of the central kernel shows that every

atom is diagonal-null. The property passes to an atomic  $\ell_1$  sum: a finite initial sum is diagonal-null, while the tail is uniformly small in operator norm because each atom has operator norm at most one. Thus  $G(e_i, e_i, e_i) \rightarrow 0$ .

Now let  $D \in \mathcal{L}_{d,p}(^3\ell_2; \mathbb{K})$ . Suppose that  $|D(e_i, e_i, e_i)| \geq \varepsilon$  on infinitely many indices. Choose  $m$  of them. The  $p$ -domination inequality yields

$$(6) \quad \varepsilon m^{3/p} \leq \|D\|_{d,p} \left( \sup_{\|a\|_2 \leq 1} \left( \sum_{i=1}^m |a_i|^p \right)^{1/p} \right)^3.$$

The supremum in parentheses is exactly

$$\begin{cases} m^{1/p-1/2}, & 1 \leq p < 2, \\ 1, & p \geq 2. \end{cases}$$

For  $p < 2$ , (6) would say  $\varepsilon m^{3/p} \leq \|D\|_{d,p} m^{3/p-3/2}$ ; for  $p \geq 2$  its right side is constant. Either is impossible as  $m \rightarrow \infty$ . Therefore  $D$  is diagonal-null. The conclusion follows for every  $T = D + G$ .  $\square$

**Theorem 4.** *The desired family  $(\mathcal{L}_{ss,p})_{p \geq 1}$  is not minimal. More precisely, the desired family  $(\mathcal{M}_p)_{p \geq 1}$  of (5) is contractively contained in it and*

$$\mathcal{M}_p \neq \mathcal{L}_{ss,p} \quad \text{for every } p \geq 1.$$

*Proof.* Define

$$\Delta : \ell_2 \times \ell_2 \times \ell_2 \longrightarrow \mathbb{K}, \quad \Delta(x, y, z) = \sum_{i=1}^{\infty} x_i y_i z_i.$$

By Cauchy–Schwarz and  $\|z\|_{\infty} \leq \|z\|_2$ ,  $\Delta$  is bounded and  $\|\Delta\| = 1$ . Every scalar-valued multilinear form  $\Phi$  is strongly  $p$ -summing with  $\|\Phi\|_{ss,p} = \|\Phi\|$ : in (1), the defining supremum contains  $\Phi/\|\Phi\|$ , while the one-point case gives the reverse norm inequality. Consequently  $\Delta \in \mathcal{L}_{ss,p}(^3\ell_2; \mathbb{K})$  for every  $p \geq 1$ .

But  $\Delta(e_i, e_i, e_i) = 1$  for every  $i$ , whereas Lemma 3 says that every member of  $\mathcal{M}_p(^3\ell_2; \mathbb{K})$  is diagonal-null. Thus  $\Delta \notin \mathcal{M}_p$  for every  $p$ . Proposition 2 supplies a proper, contractively contained desired family, which is exactly the negation of minimality in the definition of [1].  $\square$

#### VERIFICATION, LIMITATIONS, AND NOVELTY STATUS

The argument is analytic; no finite computation is being used as proof. The completeness point is explicit:  $\mathcal{G}_p$  is an  $\ell_1$  quotient and  $\mathcal{M}_p$  is a Banach direct-sum quotient. The diagonal-null property is proved after completion, not only for finite atomic sums. The norm-one inclusion  $\mathcal{M}_p \subseteq \mathcal{L}_{ss,p}$  is also explicit, so the construction meets the order convention in the source definition of minimality.

Before promotion, the run’s cheap indexes were searched for arXiv:1009.4807 and the core terminology. A bounded literature search on 22 July 2026 used

the exact title, Problem 7.2, “strongly  $p$ -summing” with “minimal”, “desired family”, and notation variants. Later papers on strongly multiple inclusions, factorable strongly summing maps, and tensor-norm summability were also checked at the title/abstract or full-text level when locally available. No explicit answer to this minimality question and no matching atomic-hull construction was found. This supports novelty only within the stated search bounds; it is not a claim of publication priority.

The theorem does not decide whether  $(\mathcal{L}_{ss,p})_{p \geq 1}$  is maximal, and it does not answer Problems 7.1, 7.3, or 7.4 of [1].

#### HUMAN-REVIEW RECOMMENDATION

The candidate is recommended for expert review as a likely valid partial solution. The highest-value checks are the atomic quotient’s csm/cud stability, the normalization and contractive inclusions, and the use of the dominated norm in (6). Pitt’s theorem and the weak- $p$  norm calculation then make the separating example short and rigid.

#### REFERENCES

- [1] D. Pellegrino and J. Santos, *Absolutely summing multilinear operators: a panorama*, Quaest. Math. **34** (2011), 447–478; arXiv:1009.4807, doi:10.2989/16073606.2011.640459.
- [2] H. R. Pitt, *A note on bilinear forms*, J. London Math. Soc. **s1-11** (1936), 174–180, doi:10.1112/jlms/s1-11.3.174.